

The diagram illustrates the logical construction of a system, organized into seven main layers, each with specific components and their roles:

- EXPERIENCE CHANNELS - ROLE AND ORGANISATION CONTEXT**
 - S** App Shell + MFE: shipper + transporter workspaces
 - O** Operator + tenant admin: control tower, finance, policy
 - M** Driver mobile: tasks, tracking + evidence
 - A** Partner API consumers: integrations + report extraction
- EXTERNAL DEPENDENCIES - CONTRACTS**
 - I** Identity federation: WorkOS + enterprise IdPs
 - E** Operational + enterprise: telematics, ERP, maps + payments, notifications + regulatory
- NATS TRUST - OPERATOR TO CREDENTIALS**
 - O** Operator: trust root
 - T** Tenant account: one per market
 - R** Org roles: RBAC + ABAC
 - C** Short-lived creds: users + workloads
- APPLICATION AND INTEGRATION BOUNDARY - POLICY-ENFORCED INGRESS**
 - E** Regional edge + WAF: TLS, routing, rate limits, country endpoint policy
 - I** Identity + policy broker: membership, RBAC + ABAC, tenant + org context
 - B** BFF + API facade: role-shaped commands + reads, no direct database access
 - R** Reporting + data API: scoped extracts + async export, masking + audit
 - A** Integration adapters: webhooks, polling + events, anti-corruption boundary
- CONTROL PLANE - LINEBOOKER TECH PLATFORM AUTHORITY**
 - T** Tenant + organisation: lifecycle + membership, authority boundaries
 - I** Identity + membership: roles + policy attributes, Linebooker authorization
 - N** NATS provisioning: accounts + credentials, claims + placement
 - R** Reference + config: country + market policy, currency defaults
 - E** Entitlements + admin: plans, billing + support, operator controls
 - O** Observe + audit: health + immutable record, masked regional telemetry
- DOMAIN SERVICE PLANE - INDEPENDENTLY DEPLOYABLE BUSINESS OWNERS**
 - M** Marketplace + tender: demand, bids, awards, owns tender state
 - L** Loads + trips + routes: allocation + execution, owns movement state
 - F** Fleet + tracking: assets, drivers, ETA, owns capacity + position
 - D** Documents: evidence + compliance, owns metadata + policy
 - D** Disputes + acceptance: proof gates + exceptions, owns dispute state
 - B** Billing + settlement: rating, invoice, payment, owns commercial state
- PLATFORM FOUNDATIONS - MESSAGING, WORKFLOW AND DISTINCT DATA RESPONSIBILITIES**
 - N** NATS Core + JetStream: RPC, commands + durable facts, tenant account boundary
 - W** Temporal workflows: long-running coordination, services retain business rules
 - P** Service-owned Postgres: transactional source of truth, database per service owner
 - Q** Projections + NATS KV: read models + fast lookup, derived, rebuildable state
 - S** Document store adapter: S3-compatible evidence store, large files outside NATS
 - S** Security, residency + immutable audit: tenant separation · RBAC/ABAC · TLS · encryption at rest · masked logs · country policy, durable business history without requiring universal event-sourced reconstruction

Arrows indicate the flow of data and dependencies between these layers, such as "authenticated requests" from Experience Channels to the Application Boundary, and "signed tenant + org context" from NATs Trust to the Application Boundary.

Logical construction. The control plane establishes authority and policy; domain services own business behavior and transactional state; NATS distributes commands and facts; Temporal coordinates processes; storage types have distinct responsibilities. This is a logical ownership model, not a claim that every tile maps one-to-one to a deployable or database.

SCOPE	L2 explains construction. Subjects, schemas, workflow steps, permission claims and runtime sizing remain in L3/L4.
AUTH	Authentication and authorization are separate. WorkOS or an enterprise IdP proves identity; Linebooker resolves tenant, organisation, role and attributes.
TENANT	A tenant is a marketplace authority. Its NATS account is a hard messaging boundary; organisations and memberships live inside that authority.
CONTROL	The control plane manages shared authority. It provisions tenants, identities, claims, policy, reference data, entitlements and operational oversight.
SERVICES	Services align to business ownership. Replicas may scale independently, but each service owns its rules, transactional state and published contracts.
NATS	NATS is communication and durable fact history. Core supports live request/reply; JetStream retains selected commands and events.
TEMPORAL	Temporal coordinates long-lived work. Tender expiry, execution, proof, claims and settlement span services without moving domain ownership into workflows.
DATA	Stores are not interchangeable. PostgreSQL is transactional truth by default; JetStream is event history; KV/projections are derived reads; Temporal stores workflow state.
DOCS	Evidence uses an adapter. POD, licences and other large documents go to an S3-compatible repository selected per deployment, not NATS Object Store by default.
AUDIT	Immutable audit is broader than event sourcing. Business facts and security/administrative actions are retained tamper-evidently; event-sourced reconstruction is selective.

- **Blue boundary** — edge and entry: channels, ingress
- **Amber boundary** — external or separately governed
- **Dashed amber line** — asynchronous event or contracted feed

ABAC Attribute-based access control
API Application programming interface
BFF Backend for frontend
ERP Enterprise resource planning
ETA Estimated time of arrival
IdP Identity provider
KV Key/value store (NATS KV)
MFE Micro-frontend
NATS The messaging system; a product name, not an abbreviation

- POD** Proof of delivery (not a Kubernetes pod)
- RBAC** Role-based access control
- RPC** Remote procedure call
- S3** The object-storage API; S3-compatible means any vendor implementing it
- TLS** Transport Layer Security
- WAF** Web application firewall
- WorkOS** A commercial identity provider, used as an example

Regional Deployment Model

This view applies the logical platform to three initial placement cells. South Africa and Australia may exchange explicitly approved NATS subject classes through gateways; Botswana is a sovereign in-country cell whose regulated business data, transit, operations and recovery do not cross the national boundary.

